LESSON 9.1

Explore Relationships Between Two Variables

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can recognize when two quantities change together, and identify which one is the independent variable and which is the dependent variable.

WHY THIS MATTERS Akela planted three trays of bean seeds for a science experiment. Each tray sat in a different spot, so each got a different amount of sunlight per day. After three weeks she measured the average plant height in every tray. Two quantities in that experiment are related: change the sunlight and the height changes with it. One of them Akela chose, and the other one answered back. Today you will name which quantity is the independent one and which depends on it, and you will find pairs like this in a jogging tracker and a pizza shop.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
independent variable *Spanish: variable independiente*	The input quantity in a model, chosen or observed.	hours studied
dependent variable *Spanish: variable dependiente*	The quantity that changes in response to the independent variable.	quiz score
depends on *Spanish: depende de*	One quantity depends on another when changing the first causes the second to change.	Score depends on study time.
quantity *Spanish: cantidad*	Something that can be counted or measured.	minutes, dollars, laps
relationship *Spanish: relación*	How two quantities change together.	more time → higher score
Table of values *Spanish: Tabla de valores*	A list of matched input and output pairs.	x: 2, 4 paired with y: 14, 28

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A bike rental costs $5 for each hour. Name the variables.

1. The quantities are hours rented and total cost.
2. Cost changes when hours change.
3. “Total cost depends on hours rented” makes sense.

Answer: hours = independent; cost = dependent

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Claire burned 123 Calories on a 30-minute jog and 246 Calories on a 60-minute jog.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The independent variable is _____ because you _____ it.
- The dependent variable is _____ because it _____ .
- _____ depends on _____ .
- If the _____ changes, then the _____ changes in response.

Word bank independent · dependent · variable · quantity · depends on · input · output · cause · respond · choose · changes · relationship

WATCH OUT!

A common mistake

Naming the variables backward

LESSON 9.1

Explore Relationships Between Two Variables

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Akela planted three trays of bean seeds for a science experiment. Each tray sat in a different spot, so each got a different amount of sunlight per day. After three weeks she measured the average plant height in every tray. Two quantities in that experiment are related: change the sunlight and the height changes with it. One of them Akela chose, and the other one answered back. Today you will name which quantity is the independent one and which depends on it, and you will find pairs like this in a jogging tracker and a pizza shop.

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1 **Circle the letter of the best answer. Show how you know.**

Akela chooses how many hours of sunlight each tray gets. Which is the INDEPENDENT variable?

(A) Plant height
(B) Hours of sunlight
(C) Number of trays
(D) Number of seeds

STEPS TO TRY
1. Which quantity did Akela get to decide?
2. Sunlight was set by where she put the tray.
3. The one subject to choice is independent.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

In Akela's experiment, which is the DEPENDENT variable?

(A) Hours of sunlight
(B) The kind of seed
(C) The size of the tray
(D) Plant height

STEPS TO TRY

1. Which quantity RESPONDED to the other?
2. Akela did not choose how tall the plants grew.
3. The responding quantity is dependent.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Claire burned 123 Calories on a 30-minute jog. Which quantity is subject to choice?

(A) Calories she burns
(B) Her heart rate
(C) Minutes she jogs
(D) The weather

STEPS TO TRY

1. What does Claire decide before she starts?
2. She sets the time on the tracker.
3. Calories are the result, not the choice.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which sentence is written in the correct order?

(A) Minutes jogged depends on Calories burned.

(B) Calories burned depends on Calories burned.

(C) Neither quantity depends on the other.

(D) Calories burned depends on minutes jogged.

STEPS TO TRY

1. Say both versions out loud. Which one is true?
2. Claire chooses the minutes.
3. The responder goes first: 'responder depends on chooser.'

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a card you and your partner put in different columns. Say the sentence 'The ____ depends on the ____' out loud both ways — which version is true, and how does that settle which variable is independent?

SENTENCE STARTER The ___ depends on the ___, so ___ is the independent variable.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help